

EMMONAK, AK, USA

WMO: 702084

Lat: 62.785N

Lon: 164.491W

Elev: 4

StdP: 101.27

Time Zone: -9.00 (NAK)

Period: 96-19

WBAN: 26650

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -31.3	(c) -28.0	(d) -33.6	(e) 0.2	(f) -27.1	(g) -32.6	(h) 0.2	(i) -26.1	(j) 16.2	(k) -7.9	(l) 14.7	(m) -4.9	(n) 1.4	(o) 10	(p) 0.856

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.3	20.2	14.8	18.0	13.5	17.0	12.9	15.3	18.8	14.4	17.5	13.3	16.3	4.2	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.7	9.8	16.5	12.6	9.1	15.7	12.0	8.8	15.1	42.7	19.0	40.1	17.3	37.4	16.1	20.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 12.7	(b) 11.3	(c) 9.9	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.3	-13.3	-11.1	-11.2	-5.5	3.2	10.2	11.9	11.2	7.3	0.0	-6.7	-12.7
	DBStd	11.06	8.56	7.85	7.21	5.91	4.68	3.22	2.52	2.65	2.98	4.04	6.06	7.96
	HDD10.0	4312	723	592	657	465	216	34	8	15	86	311	502	702
	HDD18.3	7182	982	825	915	715	470	244	198	221	330	569	752	961
	CDD10.0	171	0	0	0	0	4	41	68	52	6	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	3	0	0	0	0	0	2	0	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	4.7	5.0	5.9	5.1	5.3	4.5	4.2	4.1	4.1	4.3	4.4	4.9	4.9
(15) Precipitation	PrecAvg	619	42	43	32	27	28	37	68	93	78	67	56	46
	PrecMax	845	86	134	90	65	68	63	142	189	163	140	117	87
	PrecMin	448	12	10	8	8	10	17	19	28	20	30	9	19
	PrecStd	94	21	27	21	15	14	12	28	33	32	31	29	18
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	2.6	2.8	6.1	17.9	22.9	22.3	22.4	16.2	9.0	5.9	3.9
		MCWB	0.9	1.1	1.1	2.9	11.3	14.5	15.9	16.7	11.2	7.7	4.3	1.5
	2%	DB	2.1	1.7	2.1	3.9	15.2	20.0	19.9	19.1	13.8	7.5	3.0	2.3
		MCWB	0.7	0.7	0.8	1.8	9.6	13.8	14.8	14.8	10.5	5.7	1.9	1.0
	5%	DB	0.8	1.1	1.0	2.7	12.6	17.7	17.8	17.3	12.6	6.3	2.4	1.0
		MCWB	-0.2	0.2	0.1	1.1	8.5	12.6	13.9	13.7	10.1	4.4	1.4	0.1
	10%	DB	-2.0	-0.1	-0.9	2.1	10.4	16.1	16.4	15.1	11.4	5.1	1.2	-2.0
		MCWB	-2.7	-0.7	-1.4	0.8	7.4	11.8	13.2	12.5	9.4	3.5	0.4	-2.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	1.5	1.6	3.5	11.8	15.6	16.5	17.9	11.9	8.1	4.5	2.3
		MCDB	3.0	2.1	2.4	5.4	17.0	20.5	20.5	20.7	14.2	8.6	5.7	3.4
	2%	WB	0.6	0.7	0.7	2.1	10.2	14.3	15.3	15.6	11.0	6.0	2.7	0.9
		MCDB	1.8	1.2	1.4	3.6	14.5	18.6	18.6	18.6	13.2	7.4	3.3	2.0
	5%	WB	-0.3	0.2	0.0	1.3	8.8	13.1	14.5	14.2	10.3	4.9	1.2	-0.1
		MCDB	0.6	0.9	0.7	2.6	12.6	17.2	17.5	16.8	12.2	6.6	2.0	0.9
	10%	WB	-2.3	-0.9	-1.3	0.6	7.4	12.0	13.5	12.9	9.7	3.9	0.2	-2.3
		MCDB	-2.0	-0.3	-0.8	1.7	10.2	15.5	16.3	15.1	11.3	5.3	1.1	-1.9
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	7.0	7.4	7.1	6.5	6.7	5.3	5.1	5.1	4.5	5.2	6.3
		MCDBR	5.8	6.1	5.9	6.6	10.7	10.2	8.7	8.5	7.0	5.4	3.9	5.6
		MCWBR	5.2	5.5	5.2	5.3	6.4	5.3	4.9	5.0	4.4	4.2	3.6	5.0
	5% WB	MCDBR	5.8	5.9	5.6	6.1	10.3	9.6	8.1	8.0	6.0	4.9	4.0	5.6
		MCWBR	5.3	5.7	5.0	5.0	6.2	5.2	4.8	5.0	4.1	4.0	3.9	5.1
(40) Clear-Sky Solar Irradiance	taub		0.205	0.273	0.280	0.349	0.372	0.366	0.370	0.396	0.328	0.292	0.227	0.196
	taud		1.999	2.125	2.203	2.052	2.208	2.404	2.435	2.309	2.507	2.382	1.998	1.886
	Ebn at Noon		451	662	826	820	839	856	842	774	780	657	387	223
	Edh at Noon		33	66	90	131	123	104	98	103	69	52	32	18
(44) All-Sky Solar Radiation	RadAvg		0.24	0.95	2.69	4.51	5.24	5.28	4.19	3.35	2.14	0.99	0.27	0.10
	RadStd		0.04	0.12	0.30	0.32	0.54	0.54	0.22	0.42	0.19	0.09	0.05	0.01

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	+0.83	N/A	N/A	N/A	N/A	+0.48	+0.55	N/A	N/A	+35

Nomenclature: See separate page